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1. Prove that every connected simple graph which is not itself a tree must have at least three different spanning trees.

Solution: Let G be a connected simple graph which is not a tree. Since G is not a tree, there must be an edge xy in G that is part of a cycle. This cycle must contain at least three edges. To build a spanning tree of G , we replace this entire cycle by a single vertex, with all edges connected to vertices in that cycle connected to the vertex instead. We then build a spanning tree of the resulting graph, and then replace the vertex with the cycle again. To get a spanning tree for G , we just have to remove one edge from the cycle to make our spanning graph a tree; we have at least three choices of which edge to remove.

2. Define the "n-dimensional hypercube" graph $Q_n = (V_n, E_n)$ as

$V_n = \{\text{"Binary sequences of } n \text{ elements," } v = (v_1, \dots, v_n) | v_i \in \{0, 1\}\}$

$E_n = \{\{a, b\} \subseteq V | a \text{ and } b \text{ differ in exactly one bit, i.e. there exists a unique } i \in \{1, \dots, n\} \text{ such that } a_i \neq b_i\}.$

Let S_n be the set of all spanning trees of Q_n . We have shown in a homework that Q_n is connected for each $n \geq 1$ and since every connected graph has a spanning tree we must have that S_n is a non-empty set. It follows that $C_n = \{\text{"The number of leaves in the tree } T" : T \in S_n\}$ is a non-empty set of integers. Furthermore this set is bounded above by $c \leq |V_n|$ for all $c \in C_n$ and for each n and so each C_n will always have a unique maximal element.

Define the function $f : \mathbb{N} \rightarrow \mathbb{N}$ by letting $f(n)$ equal the unique maximal element of C_n for the graph Q_n as defined above.

Prove or disprove that $f(n) \geq 2^{n-1}$ for every $n \in \mathbb{N}$.

Solution: Let $n \in \mathbb{N}$ be fixed. If $n = 1$ then Q_1 is its own only spanning tree with 2 leaves and so $f(1) = 2 \geq 2^{1-1} = 1$. If $n \geq 2$, then view Q_n as two copies of Q_{n-1} with edges connecting identical vertices between the copies, i.e. viewing the first $n-1$ elements of the binary strings as elements of Q_{n-1} and the last element creating two copies with an edge across the copy only if the first $n-1$ values are the same. In this way, we know that Q_{n-1} is connected so has spanning tree, and so we can construct a spanning tree of Q_n by creating one in the first copy of Q_{n-1} and then adding all of the edges connecting two elements which differ in the last bit. We may check that this is indeed spanning, and a tree, and each vertex in the second copy of Q_{n-1} is a leaf, and so $2^{n-1} \in C_n$. Thus as a maximum we have that $f(n) \geq 2^{n-1}$ for every $n \in \mathbb{N}$.